

Dev Pathak

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Summary

Curious and motivated software engineer with hands-on experience building full-stack applications, automated systems, and AI-powered platforms across web development and real-time user platforms. Skilled in designing and implementing end-to-end solutions, integrating machine learning where it adds value, with a strong foundation in data structures, system design, and problem-solving. Passionate about building practical, scalable software that solves real-world problems.

Education

2026 - present	Master's (CS) at University of Massachusetts Amherst, US	[GPA: 3.9/4.0]
2021-2025	Bachelor's (CSE) at CHARUSAT University, India	[CGPA: 9.35/10.0]

Work Experience

AI/ML Intern [AkinoLabs], India Aug 2025- Dec 2025

- Developed a RAG-based resume retrieval system using semantic embeddings and vector search, enabling recruiters to efficiently identify top candidates by matching resumes with job descriptions.
- Built an automated Leave Application Chatbot with seamless integration into internal systems, allowing employees to submit and track leave requests via natural language.

Research Intern [Charusat Space Research & Technology Center], India Dec 2024- May 2025

- Applied deep CNNs to analyze remote satellite imagery, performing large-scale EDA and fine-tuning models for accurate image classification and insight extraction.
- Led research projects on oil spill detection and wildlife habitat mapping, delivering actionable results and detailed documentation for both initiatives.

Projects

Optimal Resume Retrieval System using RAG

- Developed an intelligent resume matching system using RAG, text embeddings and vector databases to retrieve the most relevant candidates based on semantic understanding.
- Improved recruiter efficiency and accuracy by enabling automatic shortlisting of top resumes from job descriptions.

Automated Leave Assistant Chatbot

- Built an interactive leave management chatbot that processes employee requests via natural language and integrates with HR systems.
- Enhanced efficiency and reduced manual workload by automating approvals, validations, and notifications for leave requests.

CrowdSorcerer

- Built a full-stack crowdfunding platform where creators can pitch projects and funders can browse and fund them through secure, transparent, decentralized transactions.
- Developed smart contracts to handle fund transfers, simulating real-world crowdfunding with trustless, tamper-proof transaction handling.

Skills

- **Technical Skills:** Python, JavaScript, C/C++, Java, HTML/CSS, React.js, Next.js, Node.js, Express.js; Machine Learning: Deep Learning, NLP, Computer Vision, RAG Systems, Semantic Embeddings, CNNs, Transformers, Feature Engineering; Predictive Modeling; MySQL, Oracle, MongoDB, Firebase, FAISS; Data Preprocessing & Analysis; TensorFlow, Keras, PyTorch, scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook; Chatbot Development, Automated Workflows, Real-Time User Interaction Platforms, Web Scraping
- **Professional/Soft Skills:** Problem Solving, Team Collaboration, Communication, Time Management, Adaptability, Empathy for Users, Decision-Making